2.9 Fractions That Close In on e

WORKSHEET

The question

STEP II 2021, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What is the largest value of $t(1-t)$ on $[0,1]$? What does that tell you about the size of T_n as n grows?

STAGE 1 Differentiating twice

Covers: Part (i)

Ask yourself Let $p(t) = t^n(1-t)^n = [t(1-t)]^n$. Differentiate once, using $\frac{\mathrm{d}}{\mathrm{d}t}[t(1-t)] = 1-2t$.

Pause and try Differentiate again and take out the common factor $n\,t^{n-2}(1-t)^{n-2}$. To match the required form, write $(1-2t)^2$ in terms of $t(1-t)$.

STAGE 2 Integrating by parts twice

Covers: Part (ii)

Ask yourself $p(t)$ is zero at $t = 0$ and $t = 1$. What about $p'(t)$, for $n \geq 2$? Why does that matter when you integrate $p(t)\,e^t$ by parts?

Pause and try Integrate by parts twice, so that p'' appears. Use part (i), divide by $n!$, and recognise the two integrals.

STAGE 3 The integer structure

Covers: Part (iii)

Ask yourself Work out T_0 and T_1 directly, by integration by parts.

Pause and try Both have the form $a + b\,e$ with integers a and b. Why does the recurrence from part (ii) keep that form?

STAGE 4 Small, and therefore informative

Covers: Part (iv)

Ask yourself For $0 < t < 1$, bound $t^n(1-t)^n$ and e^t. What does that give for T_n?

Pause and try Divide $0 < a_n + b_n e < \frac{e}{n!}$ by $|b_n|$. What does that say about the distance between e and $-a_n/b_n$?

What's really going on

Another proof that e is irrational

The question stops just short of a famous theorem. Suppose $e = p/q$ with p, q positive integers. Then

$$q\,T_n = q a_n + b_n p$$

is an *integer*. But $0 < q\,T_n < \dfrac{qe}{n!}$, and for large n the right-hand side is less than 1. So qT_n would be an integer strictly between 0 and 1, which is impossible. Hence e is irrational. This is the "trap an integer between 0 and 1" idea from Section 1.4, delivered by integrals instead of series. Integrals of exactly this kind, with polynomials like $t^n(1-t)^n$, were used by Charles Hermite in 1873 to prove that e is *transcendental*: it is not a root of any polynomial with integer coefficients.

Continued fractions

The fractions in the table are not just good; they are among the *best possible*. The continued fraction of e, found by Euler, has a beautiful pattern:

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots],$$

and its convergents are $2, 3, \frac{8}{3}, \frac{11}{4}, \frac{19}{7}, \frac{87}{32}, \frac{106}{39}, \frac{193}{71}, \dots, \frac{2721}{1001}, \dots$. Our $-a_n/b_n$ are exactly every third convergent. No fraction with a smaller denominator gets closer to e. Euler's pattern was itself the first proof that e is irrational, since only rational numbers have finite continued fractions.

Apéry's miracle

In 1978 Roger Apéry announced a proof that $\zeta(3) = 1 + \frac{1}{8} + \frac{1}{27} + \cdots$ is irrational, a question that had been open since Euler (see Section 2.3). His proof was so unexpected that the audience at first did not believe it. A year later Frits Beukers found a short version using integrals of the form

$$\int_0^1 \int_0^1 \frac{-\ln(xy)}{1-xy}\, P_n(x) P_n(y)\, \mathrm{d}x\, \mathrm{d}y,$$

where P_n is built from $\frac{\mathrm{d}^n}{\mathrm{d}x^n}\big(x^n(1-x)^n\big)$. These integrals are small, positive, and of the form "integer + integer $\times\, \zeta(3)$", divided by a controlled denominator: exactly the strategy of this question. Whether $\zeta(5)$, or $e + \pi$, is irrational is still unknown.

Try this next

1. **Run the recurrence.** Compute a_n and b_n for $n = 2, 3, 4$ from the recurrence and check the fractions in the table. Show that the signs of b_n alternate.

2. **A sharper bound.** Show that $0 < T_n < \dfrac{e}{4^n\, n!}$. How much does this improve the error estimate for $-a_n/b_n$?

3. **The irrationality proof.** Write out the argument in the box above carefully. Where exactly do you use that $T_n > 0$?

4. **A different weight.** HARDERRepeat the question with e^t replaced by e^{2t} (so $U_n = \int_0^1 \frac{t^n(1-t)^n}{n!} e^{2t}\, \mathrm{d}t$). Find a recurrence for U_n, show that U_n is a combination of 1 and e^2 with rational coefficients, and investigate whether the argument still proves that e^2 is irrational. (Compare Section 1.4.)

Further reading

- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). The classic short book on irrationality proofs, including Hermite's method.

- **Alfred van der Poorten, "A proof that Euler missed"**, *The Mathematical Intelligencer* 1 (1979). A lively account of Apéry's proof and its reception.
- **Julian Havil, *The Irrationals*** (Princeton University Press). A history of irrational numbers, written for a general mathematical reader.

The full guided solution is in Section 2.9:
https://beyondstep.keithmcnulty.org/ch02/e-approximations.html